

Cloud Architecture & Infrastructure

Week 5 • Day 2 • Technical Infrastructure & Modeling

TPM Academy



Day 2 Objectives

- Make the **region + AZ** choice in customer + compliance + cost terms
- Default to **managed services**; defend any self-managed exception
- Take a **multi-tenancy stance** tied to customer evidence
- Choose a **network boundary** with revisit triggers
- Add a **rough order of magnitude (ROM) cost table** — catch 2x surprises before they ship



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin TMD Section 1 on the wall
09:15 – 10:30	Block 1 + Activity 1	Region + AZ stance
10:45 – 12:00	Block 2 + Activity 2	Managed vs self per component
13:00 – 14:15	Block 3 + Activity 3	Multi-tenancy + network boundary
14:30 – 16:00	Block 4 + Activity 4 + Wrap	ROM cost + AI sanity check

Block 1 — Region + AZ

Where the customer's data lives matters



The Cloud Vocabulary Card

Term	What it means
Region	A geographic cloud location (East US)
AZ	Failure-isolated facility within a region
Multi-region	Across regions; survives whole-region failure
Managed service	Cloud provider runs it (Azure Database for PostgreSQL, Blob Storage, Service Bus)
Self-managed	You run it on raw compute
Multi-tenant	One deployment serves all customers
Single-tenant	Each customer gets their own
Edge / CDN	Static content cached close to users

The Five Decisions

1. **Region(s)** — which?
2. **Multi-AZ** — single, multi-AZ, or multi-region?
3. **Managed vs self-managed** — for each component
4. **Multi-tenancy** — shared / pooled / dedicated / hybrid
5. **Network boundary** — public, VPN, ExpressRoute

Each has a **cost dimension** and a **risk dimension**. Surface both.



Worked Region + AZ Stance

```
**Region:** East US (primary), West US 2 (warm-DR).  
- Customer base: 92% US; 8% Canada (cross-region OK; +80ms)  
- Compliance: SOC 2 audit retention requires named region  
  
**AZ stance:** Multi-AZ active-active for stateful tiers  
(Postgres primary + 2 read replicas across 3 AZs).  
  
**Trade-off – multi-region active-active vs warm-DR:**  
**Choice:** Warm-DR (Option A).  
**Why:** Active-active doubles cost and adds cross-region  
consistency complexity; multi-AZ within East US  
satisfies our 99.5% availability SLO.  
**Accepted cost:** Full East US outage breaches SLO ~30 min.  
**Revisit trigger:** Second multi-hour East US outage in  
12 months, OR availability SLO tightens to 99.9%+.
```

The warm disaster-recovery (DR) choice is judged against our availability Service Level Objective (SLO).

Activity 1 — Region + AZ

Quads • 45 minutes

- 5 min: pull customer location facts from Product Requirements Document (PRD) Section 2
- 5 min: pull compliance facts from Technical Concept Document (TCD) Section 3
- 20 min: decide region(s)
- 10 min: decide multi-AZ stance
- 5 min: capture trade-off (Week-4 template)



5 + 5 + 20 + 10 + 5

Block 2 — Managed vs Self

Default to managed; defend exceptions

? The Four-Question Test

Default: **use the managed service**. Reasons to self-manage:

1. Do failure modes match our needs?
2. Is cost reasonable at expected scale?
3. Does compliance allow it?
4. Do we have a specific reason to control this?

"We want full control" is a smell. Push: what specific decision do you need that the managed service forecloses?



The Managed/Self Table

Component	Managed option	Choice	Why
Postgres	Azure Database for PostgreSQL	Managed	Default; team ops capacity finite
Event Hubs	Azure Event Hubs	Managed	Same
Object storage	Blob Storage	Managed	Industry-standard durability
Custom AI service	None	Self-managed	No managed equivalent
Load balancer	Application Gateway	Managed	Trivial to use; standard

Activity 2 — Managed vs Self

Quads • 50 minutes

- 5 min: list each container (TCD Section 2)
- 10 min: identify managed-service option per container
- 25 min: managed vs self per container (4-question test)
- 10 min: capture the table



5 + 10 + 25 + 10

Block 3 — Tenancy + Network

Stakeholder-touching decisions



Multi-Tenancy Spectrum

Stance	Architecture	Pros	Cons
Shared	One deployment, all tenants	Cheap	Noisy neighbor
Pooled	One deployment + per-tenant limits	Cheap-ish; isolated	Tenant-aware code
Dedicated	One deployment per tenant	Strongest isolation	Expensive
Hybrid	Most pooled; some dedicated	Match cost to value	Two patterns

Most B2B SaaS at our stage: pooled with per-tenant rate limits (revisit Week 4 Day 4).



Network Boundary

Public HTTPS

Default for B2B
SaaS APIs. Standard
Transport Layer
Security (TLS);
nothing else.

Customer-VPN-only

Some enterprise
contracts; customer
comes through their
VPN tunnel.

ExpressRoute / Private Link

Large enterprise;
regulated industries;
dedicated network
paths.

Don't volunteer for VPN-only. It triples ops complexity. Only choose if a customer requires it.

Activity 3 — Tenancy + Network

Quads • 50 minutes

- 10 min: customer tenancy expectations (PRD Section 1)
- 10 min: pick stance with revisit trigger
- 25 min: network boundary stance
- 5 min: stakeholder implications (add to TCD Section 6 if missing)



10 + 10 + 25 + 5

Block 4 — Cost + AI Sanity

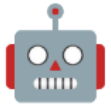
Catch 2x surprises before they ship



ROM Cost Table

Component	Stance	Cost / month	Driver
Azure Database for PostgreSQL	Multi-AZ, 2 vCore GP	400–700	Hours; storage; IOPS
Read replica	Single-AZ	200–350	Hours
Event Hubs	Standard, 5 TUs	700–1100	Throughput units; storage; egress
Blob Storage (audit)	Standard tier	\$50/TB-month	Storage; read/write ops
Application Gateway	Standard	\$25 + traffic	Hours + CUs
Egress bandwidth		50–150	\$/GB out

Goal: rough order of magnitude. Engineering will refine. The TPM job: catch the 2x surprise.



AI Sanity Check

```
Role: SRE reviewing a feature's cloud topology.  
Context: <paste TMD Section 2 sections – region, AZ, managed services,  
tenancy, network boundary, ROM cost>  
Task: Identify the top 3 risks in this topology that could  
surprise the team within 6 months.  
Constraints:  
- Be specific to the configurations described  
- For each: scenario, likelihood (L/M/H), suggested mitigation  
- Do not invent platform features  
Format: Numbered – Risk / Scenario / Likelihood / Mitigation.
```

Activity 4 — Cost + AI Sanity

Quads • 55 minutes • Wrap

- 25 min: ROM cost table
- 10 min: AI sanity check (3 risks)
- 10 min: adopt / defer / reject
- 10 min: polish Section 2 + provenance note



25 + 10 + 10 + 10 • 15-min wrap



Day 2 Wrap

What we built

- Region + AZ stance
- Managed/self table
- Multi-tenancy + network boundary
- ROM cost table
- Technical Modeling Document (TMD) Section 2 drafted

What opens tomorrow

- **REST & SOAP API fundamentals**
- Resource modeling, idempotency, errors
- When SOAP is still right

Bring to Day 3: TMD Sections 1–2 + your data model. APIs ride on top of both tomorrow.

Questions & Reflection

Where does your topology assume the team has expertise it might not have in 12 months?